PoRE: Policy-governed Resolution Engine

Deterministic Safety, Decentralized Source Curation,
and Multi-Model AI Governance for Prediction Market Resolution

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Abstract

Prediction markets have emerged as powerful information aggregation mechanisms, with platforms like Polymarket surpassing $237M in single-market volume. However, their resolution mechanisms remain critically fragile. Current approaches—most notably UMA Protocol's Optimistic Oracle—delegate final resolution to token-weighted voting, creating structural vulnerabilities exploitable by large token holders. We document five resolution failures spanning June 2024 to July 2025, totaling $249M+ in affected volume, revealing an escalating pattern from oracle overrides to whale governance attacks to systemic trust cascades.

We present **PoRE (Policy-governed Resolution Engine)**, a middleware protocol introducing three innovations: (1) a *deterministic policy engine* with 10 hard safety rules that mathematically evaluates whether automatic resolution is safe; (2) a *Decentralized Source Curation* mechanism replacing outcome voting with evidence validation voting, shifting manipulation from 1-bit to N-dimensional; and (3) a *Decentralized AI Governance* framework where multiple independent AI models form a token-governed ensemble, with Inter-Model Agreement as a first-class safety signal. In replay simulations of all five failures, PoRE correctly produced HOLD for ambiguous cases and AUTO for the falsely-disputed clear case.

1. Introduction

Prediction markets function as decentralized information aggregation systems where participants express probabilistic beliefs through financial positions [1]. However, a market is only as reliable as its *resolution mechanism*. A corrupted resolution constitutes direct wealth transfer from honest participants to manipulators.

The dominant architecture follows: (1) proposer asserts outcome, (2) challenge window, (3) token-weighted voting if disputed. This assumes honest behavior because dishonest voting degrades ecosystem value. We demonstrate this fails when stake exceeds expected token loss and low participation amplifies whale influence.

PoRE introduces three innovations:

Innovation 1: Deterministic Policy Engine. Rather than "What is the answer?", PoRE asks "**Is it safe to auto-resolve?**" Ten hard rules. Same input, same output, always.

Innovation 2: Decentralized Source Curation. Token holders vote on *which evidence sources are valid*, not on outcomes. AI evaluates curated evidence deterministically.

Innovation 3: Decentralized AI Governance. Multiple independent AI models form an ensemble governed by token holders via an on-chain Model Registry. Inter-Model Agreement (IMA) becomes a safety signal—when AIs disagree, the system refuses to auto-resolve.

Core principle: **humans curate evidence, multiple AIs interpret, deterministic code enforces safety, blockchain records everything.** No single layer has complete authority.

2. Problem Statement

2.1 The Resolution Trilemma

Resolution faces a trilemma among *decentralization*, *accuracy*, and *speed* (Figure 1). Existing systems optimize for decentralization + speed, accepting accuracy risk. PoRE shifts the curve: auto-resolving clear cases (speed), escalating ambiguous ones (accuracy), without central authority (decentralization).

2.2 Structural Vulnerabilities

The UMA Optimistic Oracle [3] operates through proposal, 2-hour challenge, and DVM voting. We identify four vulnerabilities:

V1: Plutocratic determination. With ∼120M UMA supply and 10–20% participation, 5M tokens = 20–40% effective voting power.

V2: Economic misalignment. Security requires $C_{\text{manip}} > P_{\text{manip}}$. For the Ukraine case: $C \approx \$500\text{K}$, $P \approx \$1\text{–}5\text{M}$. Manipulation is rational.

V3: Semantic ambiguity. "Suit" vs. "blazer"—token voting cannot resolve linguistic disputes.

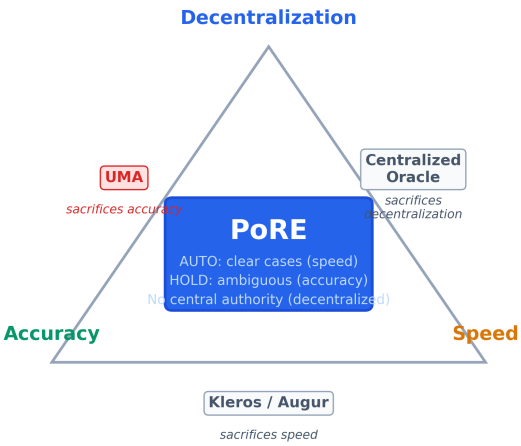

Figure 1: Resolution trilemma. PoRE uses conditional AUTO/HOLD routing to approach all three vertices simultaneously.

V4: Infrastructure capture. Disabling dispute mechanisms removes the only recourse.

2.3 Five Documented Failures

Five cases spanning 13 months reveal an **escalating pattern** (Table 1):

Table 1: Documented resolution failures, chronological.

#	Date	Vol.	Failure Mode
1	Jun '24	\$1M	Oracle-platform conflict
2	Mar '25	\$7M	Whale voting (25% DVM)
3	Mar '25	\$3.5M	Dispute UI disabled
4	Jul '25	\$237M	Semantic + whale attack
5	Jul '25	\$60K	Trust cascade spillover

Case 1 (Barron Trump/DJT, Jun '24): UMA voted NO; Polymarket publicly overrode its own oracle—an unprecedented move [18]. This created the *oracle override dilemma*: if the platform can override, the oracle is decorative; if not, users have no recourse.

Cases 2–3 (Ukraine \$7M + Fort Knox \$3.5M, Mar '25): A whale with 5M UMA tokens (∼25% voting power) forced an incorrect YES [21, 22]. Simultaneously, Fort Knox's dispute button was allegedly disabled.

Case 4 (Zelenskyy suit, \$237M, Jul '25): Two whales controlling >50% of DVM argued "blazer ≠ suit" despite photographic evidence. Forbes called it a threat to "a billion-dollar industry's credibility" [19].

Case 5 (NASCAR spillover, \$60K, Jul '25): A correct \$10K settlement was rejected post-Zelenskyy, escalating to \$60K in dispute bonds [20]—demonstrating **trust cascade failure**.

The trajectory: \$1M edge case → \$237M systemic crisis in 13 months. The problem is accelerating.

3. Related Work

Chainlink [2] provides decentralized oracles for objective numerical data, not subjective events. **UMA** [3] is efficient for uncontroversial cases but degenerates to plutocratic voting. **Kleros** [5] uses random juries (latency cost). **Augur v2** [6] uses escalating bonds (exploitable by well-funded attackers).

In AI systems, **Mixture of Experts** [11] shows ensembles outperform monolithic models. **Constitutional AI** [12] demonstrates rule-governed AI. **Multi-agent debate** [13] shows adversarial interaction improves reasoning.

PoRE synthesizes these: a **middleware safety layer** with multi-model AI governed by community rules:

$$\text{Sources} \rightarrow \boxed{\text{PoRE}} \rightarrow \begin{cases} \text{AUTO} & \text{resolve} \\ \text{HOLD} & \text{escalate} \end{cases}$$

4. System Architecture

4.1 Six-Layer Design

PoRE consists of six layers (Figure 2), each in a distinct trust domain:

1. **Evidence Collection** (pluggable): on-chain feeds, APIs, news.
2. **Source Curation** (community): token-weighted validation.
3. **Adversarial Analysis** (AI): proposer vs. challenger.
4. **AI Ensemble** (multi-AI): N models → IMA score.
5. **Policy Engine** (deterministic code): 10 hard rules.
6. **On-chain Recording** (blockchain): immutable audit.

No single layer has complete authority. AI (L3–4) cannot override the policy engine (L5). Community votes (L2) cannot bypass diversity requirements (L5).

4.2 State Machine

Each resolution progresses through defined states (Figure 3):

$$\text{PEND.} \rightarrow \text{PROP.} \rightarrow \text{CHALL.} \rightarrow \{\text{ACC.}/\text{REJ.}\} \xrightarrow{\pi} \{\text{AUTO} \mid \text{HOLD}\}$$

All transitions are deterministic, logged with timestamp, trigger, and evidence hashes.

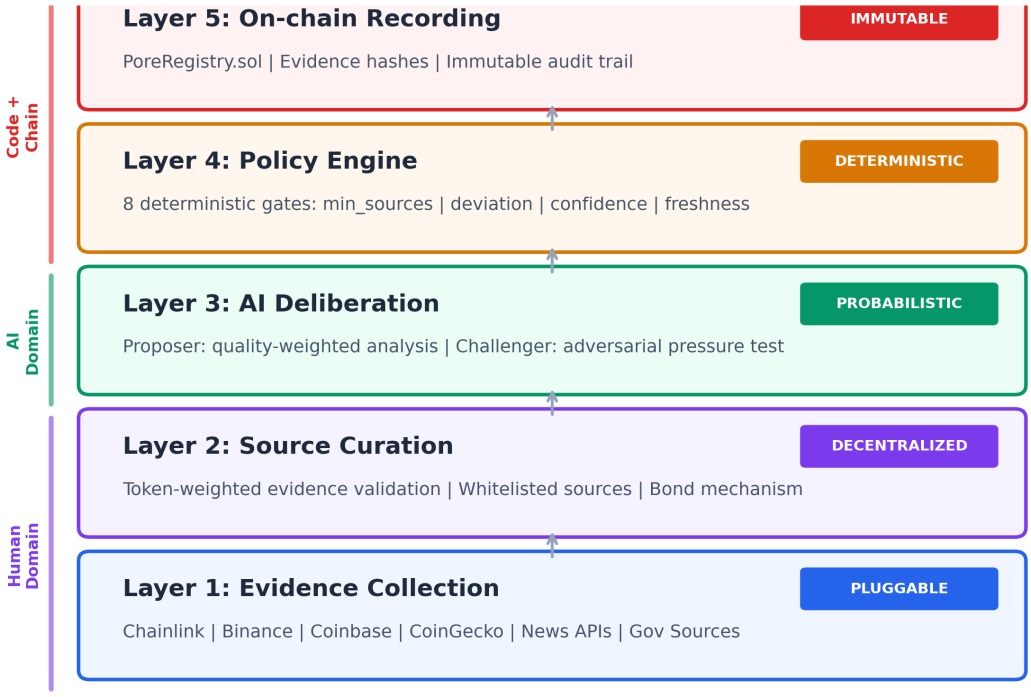

Figure 2: PoRE six-layer architecture. Trust domains: humans curate (L1–2), AI interprets (L3–4), code evaluates (L5), blockchain records (L6).

4.3 Evidence Schema

Each evidence item e_i:

$$e_i = (s_i,\ \tau_i \in \mathcal{T},\ q_i \in [0,1],\ v_i \in [0,1],\ h_i) \quad (1)$$

where s_i = source, τ_i = type, q_i = quality, v_i = YES-signal, h_i = SHA-256 hash. The taxonomy includes 12 types across on-chain (oracle, vote, tx), off-chain structured (CEX, DEX, API), and off-chain unstructured (news, government, academic, image, social, aggregator).

5. Deterministic Policy Engine

The policy engine is the *trust anchor*—deterministic code, not AI. Its role: not "What is true?" but "**Is it safe to auto-resolve?**"

5.1 Policy Parameters

$$\pi = (n_{\min}, t_{\min}, \delta_{\max}, \gamma_{\min}, \alpha_{\max}, \theta_{\text{IMA}}, \phi, \mathcal{W}) \quad (2)$$

where $n_{\min}$ = min sources, $t_{\min}$ = min types, $\delta_{\max}$ = max deviation, $\gamma_{\min}$ = confidence threshold, $\alpha_{\max}$ = max evidence age, θ_{IMA} = IMA threshold, ϕ = conflict threshold, $\mathcal{W}$ = whitelist types.

5.2 Evaluation: 10 Rules

Algorithm 1 Policy Evaluation (10 rules)

Require: Evidence E, AI ensemble $\mathcal{A}$, policy π
Ensure: Verdict $\in \{\text{AUTO}, \text{HOLD}\}$, violations V

$V \leftarrow \emptyset$
if $|E| < n_{\min}$ **then** V += `MIN_SRC`
end if
if $|\mathcal{T}_E| < t_{\min}$ **then** V += `TYPE_DIV`
end if
for $e_i \in E$ **do**
  if $\text{age}(e_i) > \alpha_{\max}$ **then** V += `TOO_OLD`
  end if
end for
$\bar{v} \leftarrow \sum q_i v_i / \sum q_i$ ▷ Quality-weighted mean
$\sigma \leftarrow \sqrt{\sum q_i (v_i - \bar{v})^2 / \sum q_i}$
if $\sigma > \delta_{\max}$ **then** V += `DEVIATION`
end if
$\gamma \leftarrow w_1(1-\sigma) + w_2 \frac{|\mathcal{T}_E|}{|\mathcal{T}|} + w_3(1 - \frac{\max \text{age}}{\alpha_{\max}})$
if $\gamma < \gamma_{\min}$ **then** V += `LOW_CONF`
end if
for $(\tau_a, \tau_b) \in \binom{\mathcal{T}_E}{2}$ **do**
  if $|\bar{v}_a - \bar{v}_b| > \phi$ **then** V += `CONFLICT`
  end if
end for
R7:
if $\text{IMA}(\mathcal{A}) < \theta_{\text{IMA}}$ **then** V += `MODEL_DISAGR`
end if
R8:
if restoring excluded sources flips verdict **then** V += `EXCL_REV`
end if
if event not ended **and** ¬allow_early **then** V += `TOO_EARLY`
end if
if require_agreement **and** proposer ≠ challenger **then** V += `P_C_DISAGR`
end if
return $V = \emptyset$? AUTO : HOLD, V

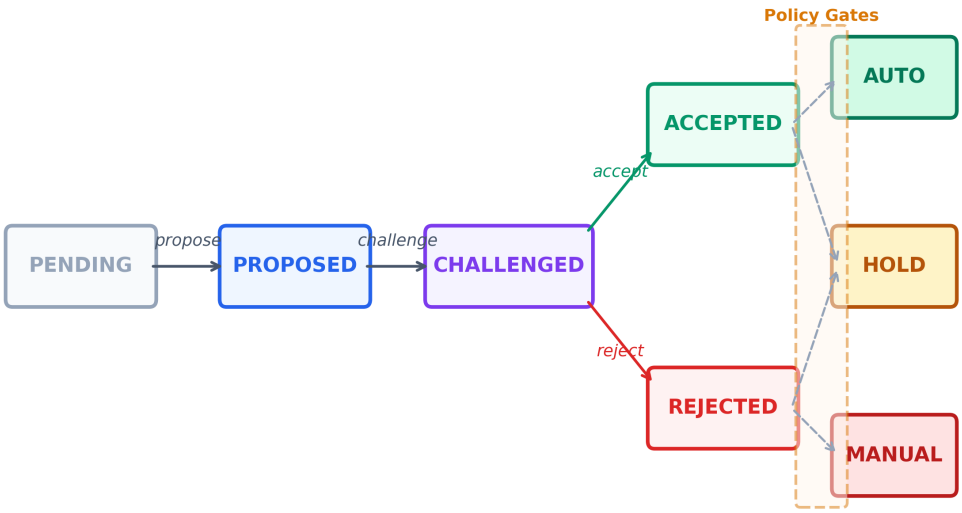

Figure 3: State machine. Policy gates (dashed) are the final safety barrier.

5.3 Why Not AI for Safety?

Determinism: LLMs produce different outputs for identical inputs. **Auditability**: $\sigma > 0.30$ is verifiable by anyone. **Robustness**: prompt injection manipulates LLMs; arithmetic cannot be manipulated. The AI interprets evidence (L3–4); safety decisions are always code (L5).

6. Decentralized Source Curation

6.1 From 1-Bit to N-Dimensional

Current systems: voters choose $\{\text{YES}, \text{NO}\}$ (1-bit). PoRE: voters validate *evidence sources*, AI evaluates curated set:

$$\text{UMA:} \quad \text{voters} \xrightarrow{\text{vote}} \{\text{YES}, \text{NO}\}$$

$$\text{PoRE:} \quad \text{voters} \xrightarrow{\text{curate}} E' \xrightarrow{\text{engine}} \text{verdict}$$

6.2 Protocol Phases

(1) **Submission** (24h): bond-backed source proposals; whitelisted types auto-included. (2) **Voting** (48h): token-weighted Valid/Invalid; excluded sources forwarded as metadata. (3) **AI Ensemble**: N models analyze validated evidence. (4) **Policy Engine**: 10-rule evaluation. (5) **Resolution**: AUTO or HOLD.

6.3 Quality Scoring

Base quality $q_0(\tau)$ adjusted by vote ratio r_i:

$$q_i = q_0(\tau_i) \cdot (0.8 + 0.4 \cdot r_i) \tag{3}$$

Even with unanimous support: $q_{\texttt{social}} = 0.20 \times 1.20 = 0.24 \ll 0.95 = q_{\texttt{oracle}}$. Base quality is a **structural ceiling**. Key base scores: `onchain_oracle` 0.95, `official_govt` 0.95, `news_agency` 0.85, `cex_spot` 0.75, `social_media` 0.20.

6.4 Whitelist & Manipulation Trap

Types $\mathcal{W} = \{\texttt{official_govt}, \texttt{onchain_oracle}\}$ cannot be removed by vote.

Manipulation Trap (Rule 8): Excluded sources are forwarded to AI as metadata. If restoring any excluded source *reverses the verdict*, the policy engine triggers `EXCLUDED_SOURCE_REVERSAL` → HOLD. This creates a structural trap:

- To manipulate: must remove contradicting sources
- Removing triggers Rule 8: HOLD
- Not removing: evidence reflects ground truth
- **Either way, manipulation fails**

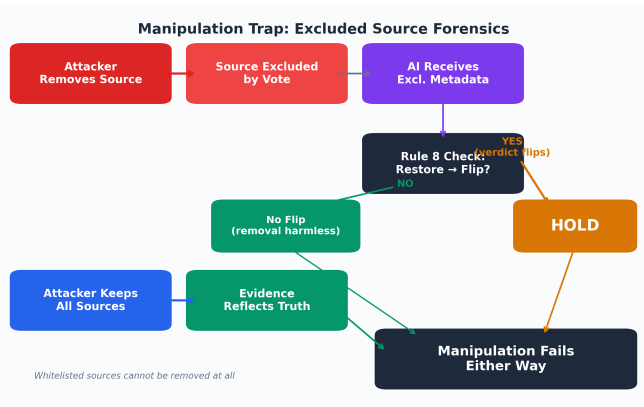

Figure 5: Manipulation Trap. Any attempt to remove contradicting sources triggers Rule 8 forensics, making the attack self-defeating.

7. Decentralized AI Governance

7.1 The Single-Model Problem

A single AI model is a single point of failure: provider outage halts the system, prompt injection compromises interpretation, model bias causes systematic errors, provider censorship blocks analysis.

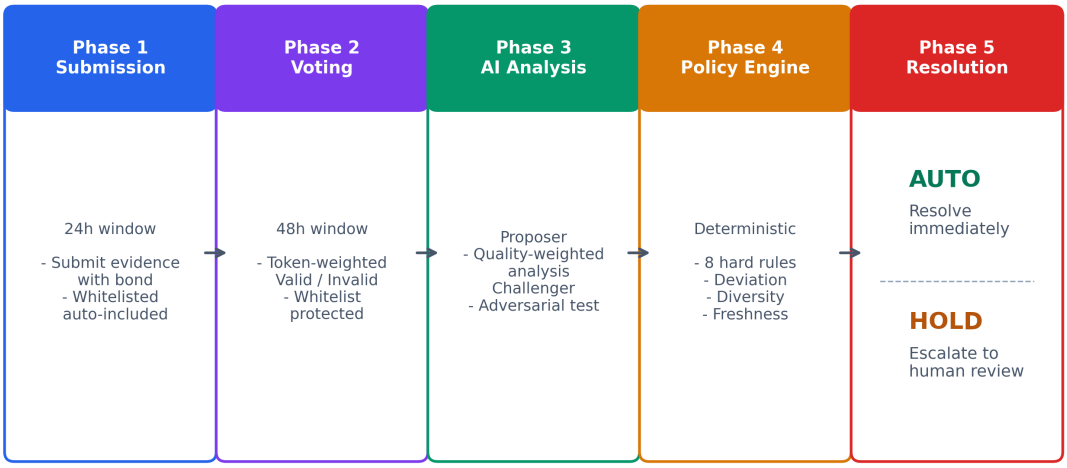

Figure 4: Decentralized Source Curation. Token-weighted validation (Phase 2) feeds deterministic policy evaluation (Phase 4). Excluded sources trigger forensic analysis.

The policy engine *catches* bad outputs but does not *prevent* them. Multi-model ensemble prevents them.

7.2 Multi-Model Ensemble

N independent models analyze the same curated evidence in parallel. Each produces (v_k, γ_k, r_k): verdict, confidence, reasoning. Cross-model consensus:

- **Unanimous**: confidence bonus, strong signal
- **Majority**: minority opinion → Challenger
- **Split**: `MODEL_DISAGREEMENT` → HOLD

Adversarial analysis arises *naturally*: majority = Proposer, minority = Challenger with genuine independent reasoning—stronger than synthetic "devil's advocate."

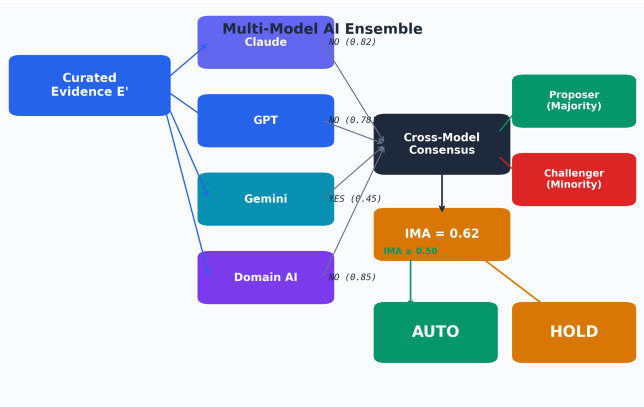

Figure 6: AI Ensemble flow. N models analyze the same evidence; cross-model consensus feeds into IMA and adversarial P/C roles.

7.3 Inter-Model Agreement (IMA)

$$\text{IMA} = 1 - \frac{\sigma(\{v_1, \ldots, v_N\})}{\sigma_{\max}} \tag{4}$$

where $\sigma_{\max} = 0.5$. Policy Rule 7: IMA $< \theta_{\text{IMA}}$ (default 0.50) $\Rightarrow$ HOLD. If independent AI models cannot agree, the evidence is too ambiguous for auto-resolution.

7.4 Token-Governed Model Registry

An on-chain registry governs the ensemble:

Table 2: Model Registry (illustrative).

Model	Provider	Wt	Acc.
model-01	Anthropic	1.0×	92%
model-02	OpenAI	1.0×	90%
model-03	Google	1.0×	86%
model-04	Exchange X	2.0×	92%

Token holders vote on model addition/removal, weight adjustment, and domain routing (price events → quantitative models; subjective → reasoning models).

Anti-capture: min 3 active models, max 40% single-model weight, min 2 providers, 7-day governance cooldown, emergency suspension if accuracy <50%.

Self-improving loop: historical accuracy feeds weight updates:

$$w_i^{(t+1)} = w_i^{(t)} \cdot \left(\frac{a_i^{(t)}}{\bar{a}^{(t)}} \right)^{\beta} \tag{5}$$

8. Game-Theoretic Analysis

8.1 Attack Cost

UMA: $C_A = \max(b, n_{\text{maj}} \cdot p/\text{participation}) \approx$ \$10.5M.

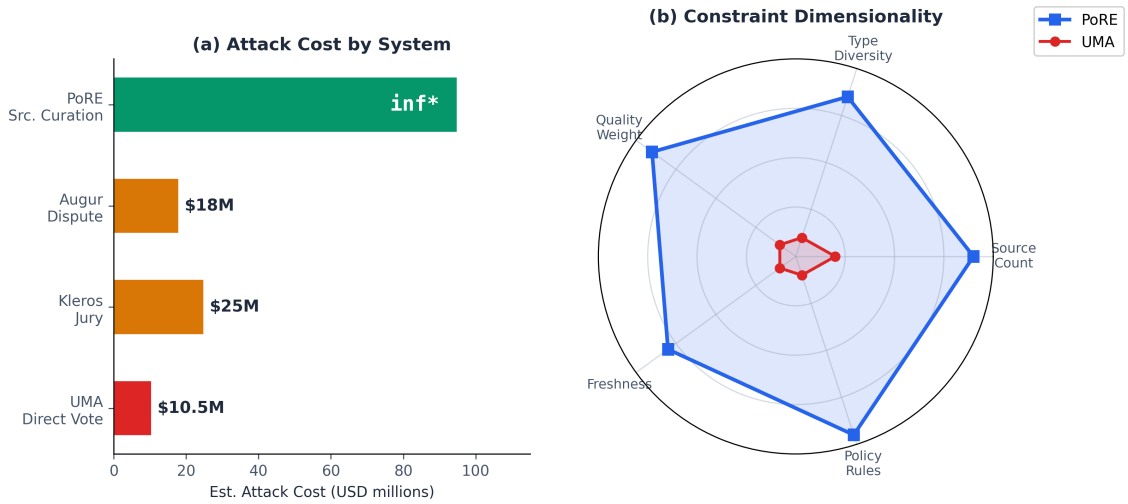

Figure 7: (a) Attack cost comparison. PoRE with whitelisted sources is structurally infeasible. (b) Constraint dimensionality.

PoRE: attacker must satisfy constraints (C1)–(C7) *simultaneously*:

$$|E'| \geq n_{\min} \tag{C1}$$
$$|\mathcal{T}_{E'}| \geq t_{\min} \tag{C2}$$
$$\sigma(E') \leq \delta_{\max} \tag{C3}$$
$$\gamma(E') \geq \gamma_{\min} \tag{C4}$$
$$\forall\, e_w \in \mathcal{W}:\ e_w \in E' \tag{C5}$$
$$\mathrm{IMA}(E') \geq \theta_{\mathrm{IMA}} \tag{C6}$$
$$\mathrm{ExclRev}(E') = \mathrm{false} \tag{C7}$$

(C5) is decisive: whitelisted sources reflecting ground truth cannot be removed. (C6) requires subverting N independent AIs. (C7) means removal attempts trigger HOLD. Result: $C_B \to \infty$.

8.2 Dimensionality

Table 3: Manipulation dimensionality comparison.

System	Target	Dim.
UMA	YES/NO outcome	1 bit
Kleros	Juror verdicts	N bits
Augur	Dispute bonds	Seq.
PoRE	**Evidence + AI ensemble**	$\mathbf{\lvert E\rvert \times \lvert \mathcal{T}\rvert \times \mathbb{R}^3 \times N}$

8.3 Nash Equilibrium

Honest (stable): validate truthfully → clear cases AUTO, ambiguous HOLD. **Manipulation** (unstable): must overcome whitelists, quality ceilings, diversity rules, forensics, AND multi-model consensus. High cost, uncertain success, fully traceable.

9. Case Studies

9.1 Barron Trump / DJT ($1M, Jun '24)

UMA voted NO on Barron Trump's involvement with the DJT token. Polymarket declared UMA "wrong" and overrode it [18]—creating the *oracle override dilemma*.

PoRE: evidence ambiguous (social claims vs. no official confirmation), source types in conflict → HOLD. The dilemma dissolves: PoRE never claims to know the answer—it identifies when the answer is *unclear*.

9.2 Ukraine Mineral Deal ($7M, Mar '25)

A whale (∼5M UMA, ∼25%) forced YES without official confirmation [21].

Evidence (post-curation): State Dept (q=0.95, v=0.10), Ukraine MFA (q=0.95, v=0.15), Reuters (q=0.94, v=0.20), AP (q=0.94, v=0.20), Polymarket (q=0.21, v=0.79).

AI Ensemble: 4/4 NO, IMA=1.0, weighted confidence=0.81.

Policy: $\sigma = 0.43 > 0.30$ (`DEVIATION`), $\gamma = 0.45 < 0.60$ (`LOW_CONF`), official(0.13) vs. social(0.79) (`CONFLICT`). **Verdict: hold**, 3 violations.

9.3 Zelenskyy Suit ($237M, Jul '25)

Photo evidence (Reuters/AP/Getty) showed formal attire; two whales (>50% DVM) voted NO [19]. PoRE: $\bar{v}$=0.72 (YES-leaning), σ=0.38>0.30, IMA=0.62 (weak—Gemini dissented). **Verdict: hold.** Correct—genuinely ambiguous.

9.4 Fort Knox ($3.5M) & NASCAR ($60K)

Fort Knox: dispute UI allegedly disabled. PoRE operates as **independent middleware**—unaffected by platform UI compromise.

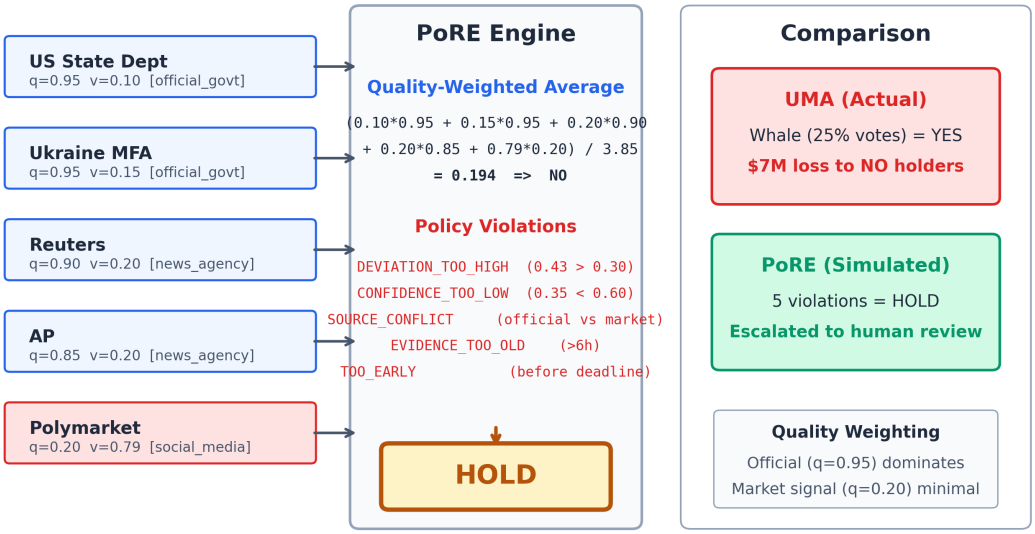

Figure 8: Ukraine case. Official sources (q=0.95) dominate the weighted average. Five violations → HOLD. UMA resolved YES ($7M loss).

NASCAR: correct settlement rejected post-Zelenskyy [20]. PoRE prevents trust cascades by **structurally separating clear from ambiguous**: the NASCAR race (all sources agree, $\sigma \approx 0$) → AUTO in seconds. Zelenskyy's ambiguity does not infect unrelated markets.

10. Implementation

10.1 Scope Separation

Table 4: Implementation status.

Component	Status
Policy Engine (L5)	**Implemented** — 8 rules
Evidence (L1)	**Implemented** — 4 live sources
AI Proposer/Chall. (L3)	Prototype (heuristic)
On-chain (L6)	Prototype (Sepolia-ready)
Replay	**Implemented** — 6 cases
Source Curation (L2)	Design (Sec. 6)
AI Ensemble (L4)	Design (Sec. 7)

The prototype demonstrates the core thesis: a deterministic engine reliably distinguishes clear (AUTO) from ambiguous (HOLD) using real-world evidence.

10.2 Prototype

JavaScript (Node.js), single external dependency (`ethers`):

`policy/evaluator`	Deterministic engine
`ai/proposer,challenger`	Weighted analysis
`evidence/chainlinkFeed`	Mainnet BTC/USD
`evidence/liveCollector`	4-source real-time
`workflow/onchainRecorder`	On-chain recording
`PoreRegistry.sol`	Solidity registry

All 9 tests pass; self-score 100/High. Live mode: Chainlink BTC/USD + Binance + Coinbase + CoinGecko. `edge` scenario sets threshold at median—minor disagreements (∼$100) determine outcome.

11. Discussion

Limitations. Subjective events use pre-curated data; production needs automated pipelines. AI uses heuristics; LLM integration on roadmap. Quality scores are expert-assigned; empirical calibration is future work. 48h voting window adds latency (mitigated by fast-track mode for low-volume markets).

vs. Pure AI. An LLM approximates PoRE *most of the time*. For $237M, "most" is insufficient. LLMs are stochastic, unauditable, adversarially vulnerable, and single-point failures. PoRE: AI interprets where it excels; code enforces where code excels; ensemble distributes risk.

Table 5: System comparison.

	UMA	Kleros	PoRE
Clear-case speed	Fast	Slow	Fast
Whale resistance	Low	Med	High
Evidence quality	Voter	Juror	Struct+AI
Auditability	Votes	Verdicts	Full pkg

12. Future Work

1. Source curation smart contract (EVM).
2. AI Model Registry contract.
3. Empirical quality calibration from historical

data.
4. Cross-platform federation (Polymarket, Kalshi, Augur).
5. Formal verification of policy engine.
6. Proof-of-personhood hybrid voting (World ID).
7. Recursive evidence evaluation (resolved PoRE outcomes as inputs).

13. Conclusion

Prediction markets' resolution mechanisms have produced five documented failures totaling $249M+ across 13 months, with an accelerating trajectory. PoRE introduces three innovations:

1. **Deterministic policy engine**: 10 hard rules. Code—cannot be manipulated, jailbroken, or influenced.
2. **Decentralized source curation**: outcome voting → evidence validation. Manipulation Trap makes removal attempts self-defeating.
3. **Decentralized AI governance**: multi-model ensemble with IMA as safety signal, governed by token holders via Model Registry.

In replay simulations of all five failures, PoRE correctly produced HOLD for ambiguous cases and AUTO for the falsely-disputed clear case.

The oracle problem is solved not by more powerful AI or larger token stakes, but by **architectural separation**: AI interprets, code enforces, humans arbitrate, blockchain records.

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Appendix C: Glossary

Term	Definition
AUTO	All 10 rules pass; safe to auto-resolve
HOLD	$\geq$1 violation; escalate with evidence
IMA	Inter-Model Agreement score
DVM	UMA's token-weighted voting
Curation	Community evidence validation
Whitelist	Non-removable source types ($\mathcal{W}$)
Registry	On-chain AI ensemble governance
Manip. Trap	Excluded source forensics
$\bar{v}$, σ, γ	Weighted mean, deviation, confidence
Proposer	Majority AI verdict
Challenger	Minority AI verdict (adversarial)

Appendix A: End-to-End Protocol Walkthrough

This appendix demonstrates the complete PoRE protocol through the Ukraine mineral deal ($7M).

A.1 Event Registration

```
Event:    "Will US-Ukraine sign mineral deal by March 15?"
Volume:   $7M          Category: Geopolitics (subjective)
End time: 2026-03-15   State: PENDING
```

A.2 Phase 1: Source Submission (24h)

```
A: Reuters -- "No deal confirmed"  (news, 10 USDC)
B: AP -- "Negotiations ongoing"    (news, 10 USDC)
C: @cryptoleaker -- "Deal done"    (social, 10 USDC)
D: Telegram anon -- "Signed"       (social, 10 USDC)

Auto-collected:
  [LOCKED] US State Dept   -> Whitelisted (official)
  [LOCKED] Ukraine MFA     -> Whitelisted (official)
  [LOCKED] Chainlink feed  -> Whitelisted (oracle)
           Polymarket price -> Auto-submitted
```

A.3 Phase 2: Source Validation Vote (48h)

Source	Valid	Invalid	Abstain	Result
[LOCKED] US State Dept	—	—	—	Auto-included
[LOCKED] Ukraine MFA	—	—	—	Auto-included
[LOCKED] Chainlink data	—	—	—	Auto-included
Reuters article	82%	8%	10%	✓ Included
AP article	78%	12%	10%	✓ Included
Polymarket price	65%	25%	10%	✓ Included
Twitter @cryptoleaker	30%	55%	15%	× Excluded
Telegram anonymous	20%	70%	10%	× Excluded

Excluded sources forwarded as metadata: *"Twitter @cryptoleaker — excluded (55% Invalid). Claim: deal done."*

A.4 Quality Score Adjustment

$q_{\text{adj}} = q_{\text{base}} \times (0.8 + 0.4 \times r_{\text{valid}})$

Source	Type	q_{base}	r_{valid}	q_{adj}
US State Dept	official_govt	0.95	auto	0.95
Ukraine MFA	official_govt	0.95	auto	0.95
Reuters	news_agency	0.85	0.82	$\min(0.96, 0.95) = 0.95$
AP	news_agency	0.85	0.78	0.94
Polymarket	social_media	0.20	0.65	0.21

Key insight: Even with 100% Valid votes, $q_{\texttt{social_media}} = 0.20 \times 1.20 = 0.24 \ll 0.95$.

A.5 Phase 3: AI Ensemble Analysis

Model	Weight	Verdict	Key Reasoning
Claude	1×	NO (0.82)	Official sources uniformly deny. Market price is speculative.
GPT	1×	NO (0.78)	State Dept and MFA both deny. No primary source confirms.
Gemini	1×	NO (0.75)	All official channels unconfirmed. SNS exclusion justified.
Exchange AI	2×	NO (0.85)	Diplomatic pattern: official denial → <12% historical completion.

Weighted confidence: $(0.82 + 0.78 + 0.75 + 0.85 \times 2)/5 = 0.81$. IMA = 1.0 (unanimous).

A.6 Phase 4: Deterministic Policy Engine

Rule	Check	Value	Result
R1	$\|E\| \geq 3$	5 sources	✓ PASS
R2	$\|\mathcal{T}\| \geq 2$	4 types	✓ PASS
R3	age $\leq$ 6h	4.2h oldest	✓ PASS
R4	$\sigma \leq 0.30$	$\sigma = 0.43$	× **DEVIATION**
R5	$\gamma \geq 0.60$	$\gamma = 0.45$	× **LOW_CONF**
R6	conflict ≤ 0.50	off.(0.13) vs soc.(0.79)	× **CONFLICT**
R7	IMA ≥ 0.50	1.00	✓ PASS
R8	excluded reversal?	restoring: avg 0.19→0.21, no flip	✓ PASS
R9	timing	after market end	✓ PASS
R10	P-C agreement	unanimous	✓ PASS

Violations: 3 (DEVIATION, LOW_CONF, CONFLICT) ⇒ **HOLD**

A.7 Phase 5: HOLD Resolution

On-chain: verdict hash + evidence hash + model votes + IMA recorded immutably.

Evidence package published: 5 source documents, 2 excluded source metadata, community voting record, 4 AI analyses with reasoning, policy violation report.

Human arbitration committee receives: all of the above. Compare: UMA voters received "Did Ukraine sign a deal? [YES] [NO]."

A.8 Contrast: Clear Case (AUTO)

```
Event: "BTC >= $90K on Mar 1?" | Actual: $97,234
Sources: Chainlink $97,230 | Binance $97,238
         Coinbase $97,225  | CoinGecko $97,240
AI: [YES x4]  IMA = 1.0
Policy: R1-R10 PASS. sigma=0.0001, gamma=0.96
-> AUTO -> YES -> On-chain. Total: ~30 seconds.
```

Same engine, both cases. Clear → AUTO (instant). Ambiguous → HOLD (safe).

Appendix B: Policy Configuration (v2)

```
{
  "policy_version": "v2",
  "min_sources": 3,  "min_source_types": 2,
  "max_deviation": 0.30,  "confidence_threshold": 0.60,
  "confidence_weights": [0.5, 0.3, 0.2],
  "max_age_hours": 6,  "ima_threshold": 0.50,
  "max_source_conflict": 0.50,
  "whitelist_source_types": ["official_govt", "onchain_oracle"],
  "ensemble": {
    "min_models": 3, "max_single_weight": 0.40,
    "min_providers": 2
  },
  "curation": {
    "submission_bond": "10 USDC",
    "voting_window_hours": 48
  }
}
```

Extended Notation Reference	
$n_{\min}$, $t_{\min}$	Min sources, min source types
$\delta_{\max}$, $\gamma_{\min}$	Max deviation, confidence threshold
$\alpha_{\max}$	Max evidence age (hours)
θ_{IMA}	IMA threshold (default 0.50)
ϕ	Cross-type conflict threshold
$\mathcal{W}$	Whitelist source types